

23/24 Sem 2 Q4(b)

Prove $(\underline{u} - \underline{v}) \times (\underline{u} + \underline{v}) = 2(\underline{u} \times \underline{v})$

Solution $(\underline{u} - \underline{v}) \times (\underline{u} + \underline{v})$
 $= (\underline{u} - \underline{v}) \times \underline{u} + (\underline{u} - \underline{v}) \times \underline{v}$
 $= \underline{u} \times \underline{u} - \underline{v} \times \underline{u} + \underline{u} \times \underline{v} - \underline{v} \times \underline{v}$

$$= 0 - (\underline{v} \times \underline{u}) + \underline{u} \times \underline{v} - 0$$

$$(\because \underline{u} \times \underline{u} = 0 = \underline{v} \times \underline{v})$$

$$= \underline{u} \times \underline{v} + \underline{u} \times \underline{v} \left(\because -(\underline{v} \times \underline{u}) \right. \\ \left. = \underline{u} \times \underline{v} \right)$$
$$= 2(\underline{u} \times \underline{v})$$